

For this task, I approached the research by first identifying the product as a self-adhesive door and wall hook and then reviewing similar adhesive-mounted hooks to understand their typical features, installation methods, and limitations. I focused particularly on the type of surfaces where the product can be safely applied, since this is important information that customers would need when using the product. Similar products using 3M adhesive systems are generally intended for clean, dry, and smooth surfaces such as painted walls, tile, glass, metal, and finished wood, while surfaces such as wallpaper, brick, rough or textured walls, fabric, and porous or unfinished materials are generally unsuitable. I also researched common installation practices, including cleaning and drying the surface before application, firmly pressing the adhesive onto the surface, and allowing sufficient time for the adhesive to bond before use. Based on this research, I would incorporate practical information such as recommended surfaces, surfaces to avoid, installation steps, product dimensions, and usage precautions into the packaging. This approach ensures that the packaging is not only visually appealing but also provides clear and useful information that helps customers use the product correctly and avoid improper application.

CTN optimization

Each retail box holds 6 units, with approximate dimensions of $27 \times 17 \times 3.55$ cm. Based on the maximum allowance of 222 units per carton, this would theoretically allow up to 37 retail boxes per carton ($222 \div 6 = 37$). However, 37 is a prime number, meaning the boxes cannot be arranged into an efficient rectangular grid. The only whole-number arrangement would be a 1×37 configuration, resulting in a carton approximately $32 \times 22 \times 129$ cm. While this configuration reaches the maximum unit capacity, it would create an excessively tall and unstable carton that would be impractical for handling, storage, and palletization.

To achieve a more practical and efficient carton configuration, I would recommend a carton size of approximately $66 \times 23 \times 32$ cm (W $\times$ D $\times$ H). This configuration can accommodate 18 retail boxes, or 108 units per carton. With 25 cartons per pallet, this results in a total of 2,700 units per pallet. This approach provides a better balance between carton capacity, stability, handling, and pallet-space utilization, rather than simply maximizing the number of units per carton.